

NITISH KUMAR

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ACADEMIC DETAILS

Indian Institute of Technology Delhi
M.Tech in Computer Science & Engineering

New Delhi, India
2025 – Present

Tezpur University
B.Tech in Computer Science & Engineering

Tezpur, India
2019 – 2023

INTERNSHIPS

Cerify | *Parallelizing the SKLEE Smart-Contract Analysis Runner*

May 2026 – July 2026

- Parallelized a sequential **KLEE**-based smart-contract analyzer into a two-phase runner with **semaphore-gated concurrency**, enabling controlled parallel symbolic execution and achieving **~2.47x speedup** with identical results.
- Fixed two concurrency bugs surfacing only under parallel execution – a **compiler-state race** and colliding output paths – validating exact equivalence with sequential output across the full regression suite.

IIT DELHI THESIS

Title: TDP-Based Distributed Symbolic Execution Engine with Global Constraint Cache on KLEE 3.2

Description: Designing a distributed symbolic execution engine on **KLEE 3.2**, implementing **test-depth partitioning** with peer-to-peer **Chase-Lev work-stealing** for decentralized task distribution, dynamic load balancing, and reduced coordination overhead, and adding a shared **GreenTrie constraint cache** that reduces the dominant SMT solver cost (**~90% of runtime**) by exploiting overlapping queries across adjacent path regions – cache hit rate rises with worker count, targeting scalable path exploration across distributed workers.

PROJECTS

Work-Stealing Parallel Runtime System

[GitHub](#)

- Built a task-parallel work-stealing runtime from scratch – custom **user-level threading**, lock-free **Chase-Lev deque**, futex-based synchronization primitives, and a fork-join API (spawn/sync, parallelfor, parallelreduce).
- Verified runtime correctness with deterministic invariant checks and **ThreadSanitizer (0 data races)**; benchmarked against OpenMP, demonstrating competitive scaling and **~95% load balance** under adversarial task-cost skew.

Integrated Language Toolchain – Shell, Compiler, VM, Debugger & GC

[GitHub](#)

- Built an end-to-end **C++ language toolchain** spanning **Flex/Bison**, AST construction, bytecode compilation, and a stack-based VM, with a process-manager shell for lifecycle management and execution control.
- Implemented an interactive debugger and **mark-and-sweep garbage collector**, supporting **breakpoints, single-step execution, variable inspection**, and per-program memory tracking, with end-to-end validation.

DB55-Memory-Constrained TPC-H Query Engine

[GitHub](#)

- Built an out-of-core **TPC-H query execution engine** under a **64MB virtual memory limit** and **10GB scratch-disk budget**, using a pipe-based disk simulator for persistent I/O and adaptive memory partitioning with buffer-pool management; ranked **top 5 in the class** on the I/O + CPU time leaderboard.
- Designed adaptive memory partitioning and a buffer pool with **O(1) LRU eviction**, with sequential-scan bypass and temporary-block recycling to control memory and disk usage; implemented dynamic filtering, projection pushdown, and hybrid hash/sort-merge joins with external k-way sorting.

TECHNICAL SKILLS

Languages: C, C++, Python, SQL

Technologies/Tools: Multithreading, OpenMP, CUDA, MPI, Git, Flex/Bison, GDB, Docker, Jadx, RISC-V

POSITIONS OF RESPONSIBILITY

National Service Scheme (Student Coordinator) | Tezpur University

2020 – 2022

- Led and coordinated nearly 700 student volunteers, managing teams, stakeholder coordination, events, and large-scale community initiatives while driving planned program execution.

Unnat Bharat Abhiyan (Student Coordinator) | Tezpur University

2021 – 2022

- Coordinated initiatives across five adopted villages, managing stakeholder interactions, student teams, and program execution for community-focused development activities.

TEACHING ASSISTANTSHIP

COL106 – Data Structures and Algorithms | IIT Delhi

Jul 2025 – Dec 2025

- Led weekly lab sessions for undergraduate students, providing hands-on guidance on algorithm design and implementation.

COL1000 – Introduction to Programming | IIT Delhi

Jan 2026 – Jun 2026

- Mentored students during lab sessions by guiding them through debugging and programming challenges.